

Straight Lines, Functions and Relations

Week 4C

Overview

Straight Lines

Key Ideas - Intersection of Two Lines & Simultaneous Equations

Function and Relations

Key Ideas - Introduction to Functions and Relations

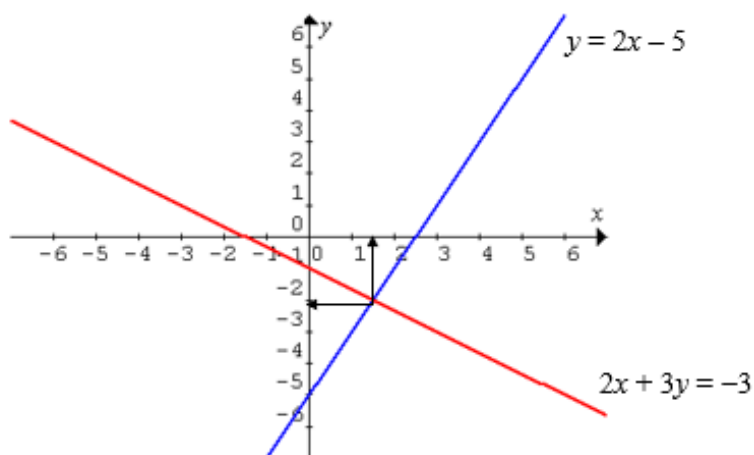
- Independent and Dependent Variables*
- Finding the Value of a Function*
- Solving for a given Function Value*
- Piecemeal Functions*

Key Idea – Intersection of Two Lines & Simultaneous Equations

Finding the Intersection Graphically

To find the intersection of the two lines, we can draw both graphs and inspect where they meet on the number plane.

Example: Find the intersection of the two lines $y = 2x - 5$ and $2x + 3y = -3$



We can see that they intersect at the point where x is about 1.5 and y about -2 . So, we estimate that the **point of intersection is (1.5, -2)**

Finding the Intersection Algebraically

We can also find the intersection of the two lines if we **solve their equations simultaneously**.

Two or more related equations that involve two or more variables are called **simultaneous equations**. We solve simultaneous equations by finding values for the variables that **satisfy all given equations at the same time**. Systems of simultaneous equations can be found in many engineering applications and in everyday problems. In this subject we will only study examples that involve two equations and two variables. At Uni you will learn to solve larger systems of simultaneous equations.

Example:

The solution to the pair of simultaneous equations

$$y = 2x - 5 \quad \text{and} \quad 2x + 3y = -3 \quad \text{is } x = 1.5, y = -2.$$

This is easily verified by substituting these values into the left-hand side of each equation to obtain the values on the right-hand side.

So $x = 1.5, y = -2$ satisfy the simultaneous equations.

Solving Simultaneous Linear Equations

Simultaneous equations can be solved either graphically or algebraically. However, algebraic methods provide exact solutions. The most common algebraic methods for solving simultaneous equations in two variables are **elimination** and **substitution**. We need to note that the graph of a linear equation is a straight line.

Method 1: Substitution

Given two linear equations with two variables that are to be solved simultaneously, the **most general method is that of substitution**:

- **Rearrange one of the equations to make one variable the subject.**
- This expression is then **substituted into the other equation**, resulting in an equation that contains a single variable.
- Once this **equation** has been **solved**, the **solution is then substituted into the “rearranged” equation** to **find** the corresponding value of the **second variable**.

***Note:** You can always check your solution by substituting the values you have found back into the original equations.*

Example:

Solve the following simultaneous linear equations:

$$y = 2x - 5 \quad (1)$$

$$2x + 3y = -3 \quad (2)$$

In this case, equation (1) is already in the required form since y is the subject.

Substitute y from equation (1) in equation (2).

$$2x + 3(2x - 5) = -3$$

$$2x + 6x - 15 = -3$$

$$8x = 12$$

$$x = 1.5$$

Now, substitute $x = 1.5$ into (1), or (2) if it is easier.

Using equation (1) we get

$$y = 2 \times 1.5 - 5 = -2$$

Hence the solution is $x = 1.5, y = -2$.

Exercise: Solve the following pair of simultaneous equations:

$$y - 2x + 5 = 0 \quad \dots (1)$$

$$5x - 2y = 4 \quad \dots (2)$$

① becomes $y = 2x - 5 \dots (3)$

sub ③ into ②

$$5x - 2(2x - 5) = 4$$

$$5x - 4x + 10 = 4$$

$$x + 10 = 4$$

$$\boxed{x = -6}$$

sub $x = -6$ into ③

$$y = 2(-6) - 5$$

$$\boxed{y = -17}$$

so $x = -6, y = -17$

Method 2: Elimination

When solving two simultaneous linear equations with two variables, **the elimination method involves eliminating one of the variables:**

- Before eliminating a variable, **re-arrange both equations** to have **only a constant on the right-hand side**.
- To eliminate a variable, it will often be necessary to **modify by multiplying one or both equations** to **obtain coefficients of the same magnitude** for one of the variables.
- Then **subtract or add one equation from the other** to **eliminate the variable** of the same magnitude.
- Once this **equation** has been **solved**, the **solution is then substituted into either of the two original equations** to **find** the corresponding value of **the second variable**.

Example:

Solve the following simultaneous linear equations:

$$y = 2x - 5 \quad (1)$$

$$2x + 3y = -3 \quad (2)$$

Firstly re-arrange equation (1) to have only a constant on the right-hand side

$$2x - y = 5 \quad (3)$$

$$2x + 3y = -3 \quad (2)$$

In these equations the coefficient of x is of same magnitude, so we don't need to modify either equation.

Add or subtract the equations to eliminate either the x or y .

In this case, subtract (2) from (3) to eliminate x .

$$2x - y = 5$$

$$2x + 3y = -3$$

$$\hline -y - 3y = 5 - (-3)$$

$$-4y = 8$$

$$y = -2$$

Now substitute $y = -2$ back into either (1) or (2). In this case I have used equation (1).

$$y = 2x - 5$$

$$-2 = 2x - 5$$

$$3 = 2x$$

$$x = 1.5$$

The solution is $x = 1.5, y = -2$.

Example:

Solve the following simultaneous linear equations:

$$2x + 3y = 13 \quad (1)$$

$$3x + 5y = 21 \quad (2)$$

Both equations are in the required form.

Modify by multiplying both equations.

Multiply (1) by 3 and (2) by 2 to give:

$$6x + 9y = 39 \quad \dots(3)$$

$$6x + 10y = 42 \quad \dots(4)$$

Subtract (3) from (4) to eliminate x .

$$6x + 9y = 39 \quad \dots(3)$$

$$\underline{6x + 10y = 42} \quad \dots(4)$$

$$10y - 9y = 42 - 39$$

$$y = 3$$

Substituting $y = 3$ back into either (1) or (2). In this case I have used (1), as it is easier.

$$2x + 3(3) = 13$$

$$2x = 4$$

$$x = 2$$

The solution is $x = 2, y = 3$.

Exercise: Solve the following simultaneous equations using elimination:

$$y = 4x - 5$$

$$5x - 2y = 4$$

$$4x - y = 5 \quad \text{--- (1)}$$

$$5x - 2y = 4 \quad \text{--- (2)}$$

$$\textcircled{1} \times 2$$

$$8x - 2y = 10 \quad \text{--- (3)}$$

$$5x - 2y = 4 \quad \text{--- (2)}$$

$$\textcircled{3} - \textcircled{2}$$

$$3x = 6$$

$$\boxed{x = 2}$$

Sub $x = 2$ into $\textcircled{1}$

$$y = 4 \times 2 - 5$$

$$\boxed{y = 3}$$

so $x = 2$, $y = 3$

Exercise: Find the equation of the straight line passing through the point of intersection of the lines $4x - 2y + 2 = 0$ and $x + 4y - 13 = 0$ and has a gradient of $\frac{2}{3}$. Give your answer in general form.

• first find the intersection point: $4x - 2y + 2 = 0$ -- (1)
 $x = -4y + 13$ -- (2)

sub (2) into (1)

$$4(-4y + 13) - 2y + 2 = 0$$

$$-16y + 52 - 2y + 2 = 0$$

$$-18y = -54$$

$$\boxed{y = 3}$$

sub $y = 3$ into (2)

$$x = -4 \times 3 + 13$$

$$\boxed{x = 1}$$

• now use $(x_1, y_1) = (1, 3)$ and $m = \frac{2}{3}$
in $y - y_1 = m(x - x_1)$

$$y-3 = \frac{2}{3}(x-1)$$

$$y-3 = \frac{2}{3}x - \frac{2}{3}$$

$$y = \frac{2}{3}x + \frac{7}{3}$$

$$\frac{2}{3}x - y + \frac{7}{3} = 0$$

($\times 3$)

$$\boxed{2x - 3y + 7 = 0}$$

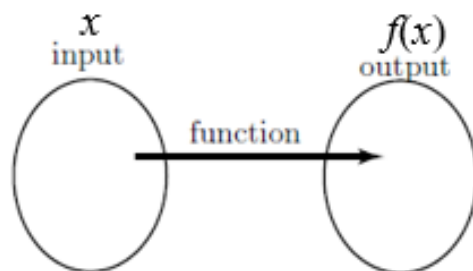
Key Idea – Introduction to Functions and Relations

Definition of a Relation:

A **relation** is a set of ordered pairs (x, y) where the variables x and y are related according to some pattern or rule.

Definition of a Function:

A function is a special **relationship between two variables**, where the value of one variable depends on the other. For our purposes, a function can be thought as a **rule that for each input there is exactly one output**. This can be show pictorially as shown below. We use two sets, one to represent the input and one to represent the output with an error showing the relationship between them.



Function Notation

The standard notation for a function is

$$f(x) = \text{a rule involving } x$$

where:

- f is the name of the function.
- x represents either a real number or variable that is put into the function and the **rule** tells what to do with the number.

We need to note that the letter of the variable is not important it is the rule that matters.

Common names for functions are f , g and h . Common variables used to express the rule are x , t and z .

Example: Given the function $y = f(x) = x^2 + x$. The letter of the variable is not important.

The rule for $f(x)$ could be written as:

$$f(\theta) = \theta^2 + \theta \quad \text{or} \quad f(t) = t^2 + t.$$

It is the rule that is important. $y = f(x) = x^2 + x$ could be described by saying:

“ The f rule takes any number – square it then adds it to itself.”

We could simply write $f(\) = (\)^2 + (\)$.

Examples of Functions:

1. $f(x) = c$ is called a **constant function**.
2. $f(x) = mx + b$ $m \neq 0$ is called a **linear function**.
3. $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$ is called a **polynomial function of degree n** .

Note: There are two special cases of polynomial functions:

- Linear Functions (polynomial of degree one ($n = 1$)).
- Quadratic Functions (polynomial of degree two ($n = 2$)).

Key Idea – Independent and Dependent Variables

The **independent** is the variable that can be chosen freely. Often it is represented by x .

The **dependent** variable depends on the independent variable and is determined by the function rule. Often it is represented by y .

Example:

Write down the independent and dependent variables of:

$$y = f(x) = x^2 + x$$

The independent variable is x and dependent variable is y .

$$h = g(t) = \frac{t^2}{2}$$

The independent variable is t and dependent variable is h .

Key Idea – Finding the Value of a Function

To evaluate a function for some real number or algebraic expression, **we need to substitute the real number** (or algebraic expression) **into the rule for every occurrence of the variable** given inside the brackets (usually x , t and z).

Examples:

If $f(x) = x^2 + 3x + 1$, find $f(-2)$

$$\begin{aligned} f(-2) &= (-2)^2 + 3(-2) + 1 \\ &= 4 - 6 + 1 \\ &= -1 \end{aligned}$$

Let $h(t) = 5 + t - \frac{3}{t^2}$, find $h\left(\frac{2}{z}\right)$.

$$\begin{aligned} h\left(\frac{2}{z}\right) &= 5 + \frac{2}{z} - \frac{3}{\left(\frac{2}{z}\right)^2} \\ &= 5 + \frac{2}{z} - \frac{3}{\frac{4}{z^2}} \\ &= 5 + \frac{2}{z} - \frac{3z^2}{4} \end{aligned}$$

Exercise:

(a) Given $f(x) = x^2 + x$ find $f(-2)$ and $f(z + y)$.

- $f(-2) = (-2)^2 + (-2) = 4 - 2 = 2$

- $f(z + y) = (z + y)^2 + (z + y) = z^2 + 2zy + y^2 + z + y$

(b) Let $f(x) = 3$, find the following $f(4)$ and $f\left(\frac{1}{w}\right)$.

- $f(4) = 3$

- $f\left(\frac{1}{w}\right) = 3$

(c) Given the functions $h(t) = 16 - t$ and $f(x) = x^2 - 4$ write down $h\left(-\frac{1}{4}\right) - f(3)$.

- $h\left(-\frac{1}{4}\right) = 16 - \left(-\frac{1}{4}\right) = \frac{65}{4}$

- $f(3) = 3^2 - 4 = 5$

- $h\left(-\frac{1}{4}\right) - f(3) = \frac{65}{4} - 5 = \frac{45}{4}$

Exercise:

Given the function $g(z) = \frac{z+1}{z^2-1}$ write down the following:

$$\begin{aligned} \text{(a) } g(x^2) &= \frac{x^2+1}{(x^2)^2-1} \\ &= \frac{x^2+1}{x^4-1} \end{aligned}$$

$$\begin{aligned} \text{(b) } g(t-1) &= \frac{(t-1)+1}{(t-1)^2-1} \\ &= \frac{t-1+1}{t^2-2t+1-1} \\ &= \frac{t}{t^2-2t} \end{aligned}$$

Exercise:

Given that $f(x) = x^2 + 3 + \frac{1}{x^2}$, show that $f(2) = f(\frac{1}{2})$

$$\begin{aligned} \bullet f(2) &= 2^2 + 3 + \frac{1}{2^2} \\ &= \frac{29}{4} \end{aligned}$$

$$\begin{aligned} \bullet f(\frac{1}{2}) &= (\frac{1}{2})^2 + 3 + \frac{1}{(\frac{1}{2})^2} \\ &= \frac{1}{2^2} + 3 + 2^2 \\ &= \frac{29}{4} \end{aligned}$$

$$\therefore f(2) = f(\frac{1}{2})$$

Key Idea – Solving for a given Function Value

When solving for a given function value, that is given the dependent value, **we need to solve the equation to find the independent value.**

It is important to be aware of the difference between substituting a value or expression (independent variable) into an equation to determine the dependent variable.

Example: Given $f(x) = 3x - 1$ find x when $f(x) = 2$.

$$\text{solve } 3x - 1 = 2$$

$$3x = 3$$

$$x = 1$$

Example: Given $f(x) = x^2 + 5x - 14$ find x when $f(x) = 0$.

$$x^2 + 5x - 14 = 0$$

$$(x - 2)(x + 7) = 0$$

$$x = 2 \text{ and } x = -7$$

Note: The values of x of the equation $f(x) = 0$ are referred to as the **zeroes or roots of the function** $f(x) = 0$. Geometrically, these values represent the x -intercepts of the graph of $y = f(x)$ (that is where it crosses the x -axis)

Exercise:

(a) Determine the zeroes, of the following function $f(x) = x - x^2 + 5$. Leave as exact roots.

$$x - x^2 + 5 = 0$$

$$x^2 - x - 5 = 0$$

$$\begin{aligned} a &= 1 \\ b &= -1 \\ c &= -5 \end{aligned}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-(-1) \pm \sqrt{(-1)^2 - 4 \times 1 \times (-5)}}{2 \times 1}$$

$$x = \frac{1 \pm \sqrt{21}}{2}$$

Exercise:

(b) Find the roots of the function $g(x) = \frac{7-3x}{2}$

$$\frac{7-3x}{2} = 0$$

$$7-3x = 0$$

$$7 = 3x$$

$$3x = 7$$

$$x = \frac{7}{3}$$